

Design of an Upstream Bioprocess

Student Instructions

Teams of three design the upstream (cell culture/fermentation) process for a real bioproduct. This is a design exercise, not a literature review — apply what you've learned about kinetics, material balances, oxygen transfer and reactor sizing to real numbers. Reports with no calculations, no mass balances, and no committed process design will lose marks regardless of writing quality.

1. Teams and Product Choice

- Fixed teams of 3. No exceptions without a genuine reason — talk to us if there is one.
- Pick one bioproduct: a modern biopharmaceutical (mAb, recombinant insulin, recombinant enzyme, vaccine antigen, AAV, GLP-1 peptide) or a classic fermentation product (penicillin, citric acid, lactic acid, lysine, ethanol).
- No two teams may share a product/host combination — register by the deadline, first-come, first-served. Something else in mind? Get it approved first.

2. Think Like a Biochemical Engineer: Rate, Yield, Titer, CQAs, PAT

Your job is to make the process work at scale and know exactly how well it's working. That comes down to a few numbers that should show up throughout your report:

- Titer — product per litre of broth (g/L). Determines whether downstream processing is affordable.
- Rate (volumetric productivity) — how fast you make it (g/L/h).
- Yield — how efficiently substrate becomes product (Yp/s). Drives raw material cost.
- State your predicted titer, rate, and yield explicitly in Section 3.4 — don't leave the reader to work them out.
- CQA (Critical Quality Attribute) — a product property that must stay in spec (e.g. glycosylation, potency, purity). Identify your product's CQAs and the factors that affect them.
- PAT (Process Analytical Technology) — measuring and controlling key parameters in real time so CQAs stay on target. Briefly note how you'd monitor your process.

Weave this into the sections below rather than treating it as a separate chapter. Quoting titer, rate, yield, CQAs, and a monitoring strategy is what makes a report read as engineering.

3. What Your Report Needs to Cover

The final report (Stage 2) should work through all eight items below. Where real data isn't available, state a clear assumption and calculate from it — that's expected. Numbers with no basis are not.

3.1 Product and process definition

- What you're making and why it matters; host organism/cell line and why; product characteristics, target market, key CQAs.

3.2 Biological system

- Growth vs. product formation (growth-associated, non-, or mixed); key pathways; substrates/nutrients/by-products; batch, fed-batch, or continuous, with a reason.

3.3 Bioreactor and upstream process

- Bioreactor type and scale; operating conditions (T, pH, DO, agitation, aeration); inoculum, media, feeding strategy, process duration; your PAT/monitoring approach.

3.4 Engineering calculations — the heart of the project

Weighted most heavily. At minimum, show your working for:

- A material balance over the bioreactor (substrate, biomass, product, by-products).
- Growth and product-formation kinetics with sourced or estimated parameters.
- Yield coefficients (Yx/s, Yp/s).
- Oxygen transfer: kLa, oxygen demand (OUR), agitation/aeration needed.
- Heat generation/removal, if relevant.
- Reactor volume, and your predicted titer, rate, and yield.

3.5 Process optimization

- Main bottleneck; variables that move rate/yield/titer the most; one specific, justified improvement — not a vague 'optimize conditions'.

3.6 Downstream processing (high-level only)

- Major recovery/purification steps. No equipment sizing or resin selection needed.

3.7 Rough economics

- Main cost drivers — media/feed, reactor time, oxygen/aeration, purification. Order-of-magnitude only, no CAPEX model.

3.8 Final recommendation

- The process you'd actually run and why; key assumptions; biggest risks; what you'd explore next.

4. How the Project Is Evaluated

Two stages. Stage 1 checks your project is realistic before you commit most of your time. Stage 2 is the full design and carries most of the grade.

Stage	Timing	Deliverables	Weight
1 — Progress Check	Immediately after the mid-semester exam	1–2 page write-up + 5-min team presentation	30%
2 — Final Design	Last two weeks of instruction	Up to 10-page report + 10-min team presentation	70%

4.1 Stage 1 — Progress Check (30%)

Timing: immediately after the mid-semester exam.

Submit a 1–2 page write-up: product, host, operating mode with rationale; a first-pass process outline; one calculation already underway (need not be finished or correct); open questions. Then a 5-minute team presentation (everyone talks), with quick feedback from us. Graded on scope, feasibility, and evidence, real work has started — not polish.

4.2 Stage 2 — Final Report and Presentation (70%)

Timing: the last two weeks of instruction.

Up to 10 pages (excluding cover page, references, appendices) covering all eight sections above, plus a 10-minute team presentation and Q&A.

10 pages is a ceiling, not a target — a tight, calculation-heavy 7 pages beats a padded 10. Use appendices for lengthy derivations.

5. What Earns Marks (and What Doesn't)

Earns marks	Won't earn marks
Real material balances, kinetics, k_La /oxygen-demand calculations with working shown, plus stated titer/rate/yield	Retelling a textbook or review article with no numbers of your own
Assumptions stated clearly and justified	Numbers presented as fact with no basis
A specific, quantified improvement idea	Generic advice like 'optimize the conditions'
A short, well-sequenced downstream and economics section	A long downstream/economics write-up padding a thin upstream design
A clear final call, with risks and assumptions spelled out	A report that never commits to a process or operating conditions

6. Format and Submission

- One PDF via the course portal by the deadline; usual late policy applies.
- Cover page (title, team names, host organism); cite sources for parameters/process data.
- Everyone must speak at both presentations — individual contribution and Q&A may factor into the grade.